

| <b>Department of I.C.T., Veer Narmad South Gujarat University, Surat</b><br><b>Internal Examination 2024 (Date: 21 October 2024)</b> |                                      |                     |       |                      |                  |
|--------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|---------------------|-------|----------------------|------------------|
| <b>Semester:</b>                                                                                                                     | M.Sc.(I.T.) 7 <sup>th</sup> Semester | <b>Total Marks:</b> | 40    | <b>Exam Seat No.</b> | <b>Signature</b> |
| <b>Subject:</b>                                                                                                                      | 706 - Practical 16                   | <b>Duration:</b>    | 2 Hrs |                      |                  |

- ❖ Create an enumeration for employee designation with values General Manager, Project Manager, Team Leader, Developer and Tester.
- ❖ Create a database for employee data management with Name, Age and Designation in Employee Table and Employee Increment detail with Employee, Basic Salary, Increment and New Basic Salary in Increment Detail table.
- ❖ Use LINQ wherever applicable.

**Q.1 Create an MVC Core Web API for following requirements: [10]**

- ✧ Prepare an endpoint to input Employee Designation and Basic Salary to calculate salary increment and return Increment Value from the endpoint.
- ✧ Use enumeration for employee designation.
- ✧ Salary Increment = Basic Salary X Employee Designation wise increment %
- ✧ Employee Designation wise Increment percentage is as follows:
  - General Manager 5%, Project Manager 10%, Team Leader 15%, Developer 20% and Tester 20%.

**Q.2 Create an MVC Core Web application for Salary Increment. [20]**

- ✧ Using Entity Framework, insert employee salary increment data in Increment Detail Table.
- ✧ Use enumeration for employee designation.
- ✧ Allow user to select specific employee to insert salary increment data.
- ✧ Use Web API created in Q.1 to get increment value and calculate new salary as per available data before inserting data in table. No manual data entry for Increment and New Basic Salary values.

**Q.3 Perform the following operations on given data set **laptops.csv**: [10]**

1. Display information of laptops having brand "Apple".
2. Display information of laptops having Touchscreen.
3. Display the information of laptop having Touchscreen and 8GB RAM.
4. Calculate the average price of Apple laptops.
5. Count the Model Name wise Apple laptops
6. Represent categorical data of laptops graphically for laptop Category.
7. Graphically show Influence of Category Class on laptop price.
8. Graphically show brand wise laptops count.
9. Graphically show the frequency distribution of Operating System.
10. Graphically show the contribution of all brands and explode the brand having highest contribution.